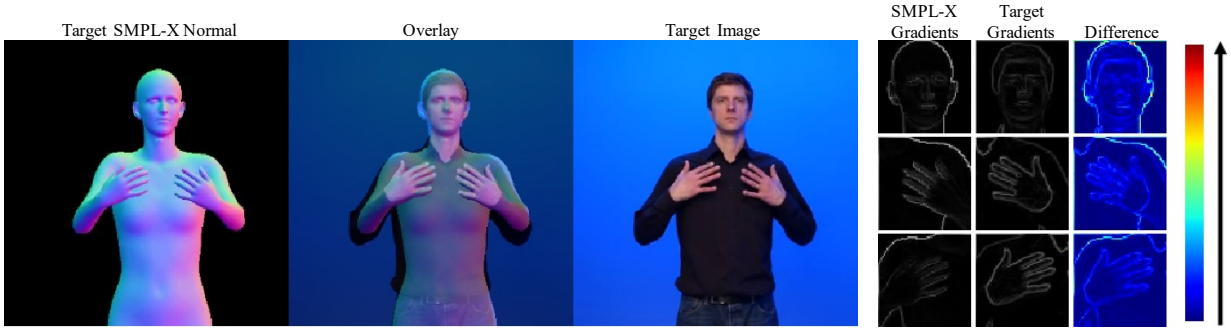
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GReg: Geometry-Aware Region Refinement for Sign Language Video Generation

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a) Shape comparison. b) Key regions edge comparison.

Figure 1. SMPL-X [20], as a volumetric human body representation, effectively captures the semantic features of the human body, such as shape. As illustrated in a), SMPL-X achieves high-quality spatial alignment with the human body. Furthermore, for key region edges, as shown in b), SMPL-X demonstrates precise alignment with the edges of the human hand, resulting in a low gradient difference. This high level of alignment significantly enhances the generation capability, enabling more accurate and realistic outputs.

Abstract

Sign Language Video Generation (SLVG) aims to transform sign language sequences into natural and fluent sign language videos. Existing SLVG methods lack geometric modeling of human anatomical structures, leading to anatomically implausible and temporally inconsistent generation. To address these challenges, we propose a novel framework: Geometry-Aware Region Refinement (GReg) for SLVG. GReg uses geometric information (such as normal maps and gradient maps) from the SMPL-X model to ensure anatomical and temporal consistency. To fully leverage the geometric priors, we propose two novel methods: 1) Regional Prior Generation, which uses regional expert networks to generate target-structured regions as generation priors; 2) Gradient-enhanced Refinement, which guides the refinement of detailed structures in key regions using gradient features. Furthermore, we enhance visual realism in key regions through adversarial training on both these regions and their gradient maps. Experimental results demonstrate that GReg achieves state-of-the-art performance with superior structural accuracy and temporal consistency.

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1. Introduction

Sign language plays a crucial role as a communication tool for the deaf and hard-of-hearing communities, offering a unique means of interaction within society. Sign Language Production (SLP), which converts natural language into sign language videos, can further bridge the communication gap between the deaf and the hearing. Typically, the SLP pipeline consists of 2 steps: Text2Sign and Sign2Video. First, natural language is transformed into a corresponding sequence of sign language gestures, and then, a realistic sign language video is generated from the sign sequence. The Text2Sign part has made significant progress, with methods like [1, 28–30, 35, 42, 47] capable of converting natural language into coherent sign language motion sequences. However, the Sign2Video part, also known as Sign Language Video Generation (SLVG), remains a significant challenge in this field.

Current SLVG approaches such as [6, 8, 16, 26, 31, 36, 37, 39], by using OpenPose [3] key points or heatmaps, suffer from a fundamental limitation: the lack of geometric information. These methods often produce results that fail to preserve the natural anatomical structure of the human body, leading to incoherent motions and unrealistic

gestures. [32] utilize optical flow to approximate motion transfer, enable temporally coherent generation but rely on real video sequences as driving conditions, and can not be applied to SLP, limiting its applicability to novel sign language video creation. Thus, we aim to address these challenges by leveraging 3D geometric information of human pose, including shape and edge, to improve the structural accuracy and temporal coherence of SLVG.

The SMPL-X model provides a parametric representation of the human body, enabling the generation of 3D meshes with body pose, hand pose, facial expression, and shape parameters. Although SMPL-X has been used as a motion representation for Text2Sign in SLP by methods such as [1, 42, 47], effectively utilizing fine-grained information from aligned details for Sign2Video remains challenging. In this work, we propose Geometry-Aware Region Refinement (GReg), which uses normal maps rendered from SMPL-X as a volumetric and anatomically consistent representation of sign language motions. As shown in Fig. 1, normal maps provide detailed geometric information for structural correctness and enable the estimation of optical flow for continuous and coherent SLVG.

Moreover, sign language understanding relies heavily on facial and hand movements, which convey most of the semantic information. However, previous methods [8, 31, 32] that model and generate the entire image directly may fail to capture fine-grained details in these critical regions. To address this, we propose a region refinement strategy with two novel techniques: Regional Prior Generation (RPG) and Gradient-enhanced Refinement (GR). RPG uses Regional Experts (RE) to generate structure priors for key regions like the face and hands. GR then refines these regions during global generation by leveraging gradient features via attention mechanisms. Finally, we employ an adversarial training framework that incorporates cropped key regions and their gradient maps to enhance structural precision.

The contributions of this work can be summarised:

- We propose a novel Geometry-Aware Region Refinement (GReg) framework for sign language video generation, which fully exploits the geometric information of SMPL-X for region-specific refinement and adversarial training.
- We introduce two key methods: Regional Prior Generation (RPG) and Gradient-enhanced Refinement (GR) which focus on incorporating local priors and refining key regions for detailed refinement.
- Our method achieves state-of-the-art performance in generating realistic, anatomically accurate, and temporally coherent sign language videos, and it can be directly integrated into SLP methods for Sign2Video.

2. Related Works

2.1. Human Pose Representation

Skeleton [3] models kinematic graph through sparse 2D/3D joint coordinates and skeletal connections , offering computational efficiency for real-time applications, such as action recognition [41], sign language recognition [5, 11, 13], and sign language translation [5]. However, when applied to SLVG, the skeleton's lack of volumetric shape modeling leads to ambiguity in representing complex gestures.

Heatmap [6, 32] encodes 2D spatial distributions of keypoints via probability maps , implicitly capturing partial volumetric cues, so it is widely used as a result representation for pose estimation [2, 18]. However, when applied to the conditions of SLVG, it is not able to explicitly express geometric constraints on anatomical structures.

SMPL-X [20] introduces a differentiable human 3D mesh with 10,475 vertices and 54 joints, modeled by shape $\beta \in \mathbb{R}^{10}$, pose $\theta \in \mathbb{R}^{75}$, hand $\phi \in \mathbb{R}^{24}$, and expression $\psi \in \mathbb{R}^{10}$ parameters. Benefiting from its expressive motion capabilities, in addition to SLP, it has been widely applied to motion generation [43] and human-object interaction [45]. It can effectively represent volumetric human structures through XYZ surface normal map $N \in \mathbb{R}^{3\times H\times W}$, while the gradient map $G \in \mathbb{R}^{H\times W}$ explicitly captures edge information, particularly highlighting key regions such as the face and hands.

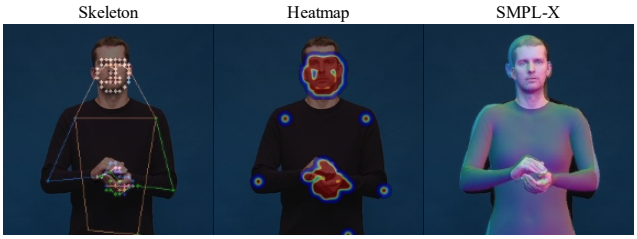


Figure 2. Comparison of different human representations

2.2. Sign Language Production

Sign Language Production (SLP) was first proposed by [27], and aims to translate spoken language into sign sequences. Current SLP pipelines [31] decompose this into Text2Sign and Sign2Video stages. While early skeleton-based methods [27, 29, 30] dominate due to motion kinematics compatibility, their abstraction of body shape them less comprehensible for sign language users. Recent SLP works [1, 42, 47] have adopted SMPL-X, which incorporates human body shape, to represent sign sequences for better comprehensibility. However, Sign2Video tasks based on SMPL-X representations have not yet been investigated.

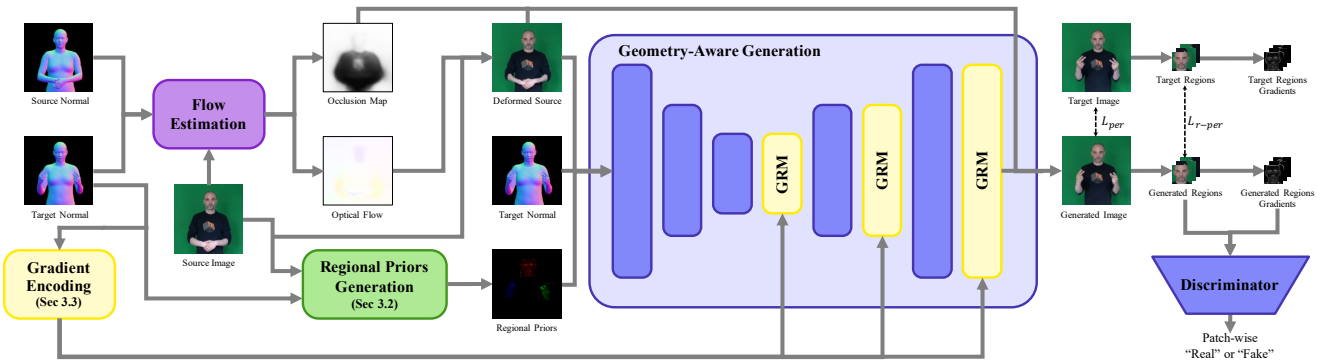


Figure 3. **Geometry-Aware Region Refinement (GReg)** includes 4 components: flow estimation, regional prior generation, gradient encoding, geometry-aware generation, and adversarial training. Its specific details are described in Sec. 3.1

2.3. Sign Language Video Generation

Sign Language Video Generation (SLVG) aims to produce detailed, realistic, and temporally consistent sign language videos from sign sequences. Existing methods [22–24] can perform human pose transfer but fail to generate details in key regions (e.g., face, hands) or ensure temporal consistency. Methods based on skeleton generation [26, 31, 36, 39] lack geometric information, leading to anatomically incorrect outputs, especially in key regions, which further results in poor temporal consistency. While [32] achieves coarse motion transfer by estimating volumetric motion from real videos and enhances details with heatmaps, it relies on real videos for optical flow and struggles with incorrect human anatomy.

3. Methodology

3.1. Overview

As shown in Fig. 3, we propose a GAN-based framework, **G**eometry-Aware **Reg**ion Refinement (**GReg**), which introduces volumetric 3D geometry from SMPL-X as control conditions to refine key regions. The pipeline operates as follows:

- **Flow estimation**: The SMPL-X normal maps of the source and target poses (N_s and N_t) are fed into an optical flow estimator to predict a dense flow field F and its corresponding occlusion mask O. The volumetric property of SMPL-X enables accurate dense flow estimation.
- **Region prior generation**: Crop K key regions (e.g., face and hands) from N_t and process them jointly with I_s through the dedicated Regional Experts to generate refined regions with structure. Then paint these regions into K blank canvases at their original positions, forming region prior maps R, which serve as structured guidance for subsequent refinement processes.
- **Gradient map encoding**: Compute gradient maps G_R for K key regions of N_t, and encode through the gradient encoder to produce a detail-enhanced feature X_g.
- **Geometry-aware generation**: Warp I_s via F to align with the target pose, yielding $I_{s\to t}$. The spatially aligned $I_{s\to t}$, target normal map N_t, and region prior maps R are concatenated and fed into the global generator. The global generator progressively refines edge details via Gradient-enhanced Refine Modules (GRM) under the control of X_g, producing an intermediate result I_{gen}. Blend $I_{s\to t}$ and I_{gen} using the occlusion mask O to obtain the final output $\hat{I}_t$.
- **Adversarial training**: The framework is optimized via adversarial training, perceptual loss, and geometric consistency constraints.

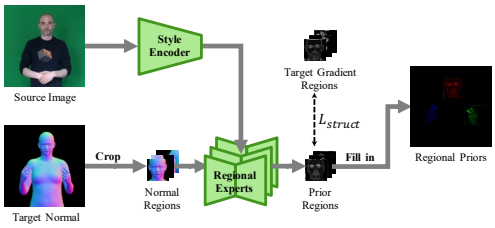


Figure 4. **Region prior generation (RPG)**. RE transforms the input normal regions into corresponding structural priors constrained by target gradients through the style encoding of the source image, and then prior regions are filled into blank canvases as the regional priors.

3.2. Regional Prior Generation

As shown in Fig. 4, to fully leverage the geometric information of key regions, we separately train Regional Experts (RE) for the face and hands to perform local generation, serving as a prior for global generation, thereby enhancing the clarity of local regions. The RE structure adopts a UNet [25] architecture that integrates modulated convolutions from StyleGAN2 [15]. These convolutions enable the modulation of style codes into spatial features.

Specifically, we encode the source image I_s to obtain the style code $X_{style} \in \mathbb{R}^{C_{style}}$, which serves as the style input for the RE. Meanwhile, we crop the face region N_f, left hand region N_{lh}, and right hand region N_{rh} from the generation condition N_t with geometric information, where $N_f, N_{lh}, N_{rh} \in \mathbb{R}^{3\times H_r\times W_r}$, forming three key regions. The face region is input into the face RE to generate R_f; the left and right-hand regions are input into hand RE to generate R_{lh} and R_{rh}, where $R_f, R_{lh}, R_{rh} \in \mathbb{R}^{H_r\times W_r}$. We constrain them using the corresponding target regional gradient $G_f, G_{lh}, G_{rh} \in \mathbb{R}^{H_r\times W_r}$, ensuring RE can fully learn structure information of the key regions. Benefiting from the specialization of each RE on a local region, they can better capture the details of key regions. Finally, generated regions R_f, R_{lh}, and R_{rh} are filled into blank canvases according to their original positions to obtain $R \in \mathbb{R}^{3\times H\times W}$, ensuring spatial alignment, which is then used as regional priors for the subsequent global generation.

3.3. Gradient-enhanced Refinement

As shown in Fig. 5, in the global generation stage, the geometry-aware generator takes spatially aligned inputs $I_{s\to t}, N_t, R$, where the Gradient-enhanced Refinement Module (GRM) utilizes features obtained through gradient encoding as input and employs a gradient-guided attention mechanism to refine the structural details of human body edges, as illustrated in Fig. 5.

Firstly, we create spatial Gaussians as the weight matrices for the key regions of the face and hands, denoted as $\mathcal{W}_f, \mathcal{W}_{lh}, \mathcal{W}_{rh} \in \mathbb{R}^{H\times W}$, these weight matrices are centered at the midpoints of their respective regions with a standard deviation of σ and their calculation is uniformly expressed as:

$$\mathcal{W}_r(i,j) = \exp\left(-\frac{(i-i_r)^2+(j-j_r)^2}{2\sigma^2}\right), \quad (1)$$

where, $r \in \{face, lh, rh\}$, (i_r, j_r) represents the midpoints coordinates of the corresponding region, and (i, j) denotes the coordinates of any point in space.

The gradient map $G_{t,n} \in \mathbb{R}^{H\times W}$ is computed via backward differences on the N_t:

$$\begin{aligned} N_t^* &= \mathcal{G}(N_t) \\ I_x(i,j) &= N_t^*(i,j) - N_t^*(i-1,j), \forall\, i \geq 2,\ j \geq 2 \\ I_y(i,j) &= N_t^*(i,j) - N_t^*(i,j-1), \forall\, i \geq 2,\ j \geq 2 \\ G_{t,n}(i,j) &= \sqrt{I_x(i,j)^2 + I_y(i,j)^2 + \epsilon}, \end{aligned} \quad (2)$$

where $\mathcal{G}(\cdot)$ is grayscale conversion, $N_t^* \in \mathbb{R}^{H\times W}$ is the grayscale-converted normal map, $\epsilon = 10^{-6}$ prevents division instability, and indices (i, j) start from the padded boundary after applying zero-padding (left and top).

In gradient encoding, we use the weight matrices $\mathcal{W}_f, \mathcal{W}_{lh}, \mathcal{W}_{rh}$ to weight the $G_{t,n}$, obtaining gradient maps $G_R \in \mathbb{R}^{3\times H\times W}$ for each key region. The G_R are calculated as:

$$G_R = [\mathcal{W}_{face} \odot G_{t,n},\ \mathcal{W}_{lh} \odot G_{t,n},\ \mathcal{W}_{rh} \odot G_{t,n}], \quad (3)$$

where $\odot$ denotes element-wise multiplication, G_R stacks three spatially weighted gradient maps along the channel dimension, and each channel $G_R^{(k)} \in \mathbb{R}^{H\times W}$ ($k \in \{face, lh, rh\}$) highlights gradient responses within the corresponding key region. These gradient maps are then encoded into gradient features X_g.

In GRM, the gradient feature X_g serves as attention Query, interacting with the GRM's input feature $X_{in} \in \mathbb{R}^{C_{in}\times H_{in}\times W_{in}}$ (Key/Value) via a cross-attention mechanism, as:

$$X_{out} = X_{in} + \text{softmax}(\frac{Q(X_g)K(X_{in})^T}{\sqrt{C_{in}}})V(X_{in}), \quad (4)$$

where, Q, K, V are convolutions that map the input features to the Query, Key, and Value spaces, respectively. Using gradient features as a query significantly enhances the edge sharpness and structural details of the generated images, particularly in key regions, thereby achieving fine-grained refinement of local details.

3.4. Training Strategy

Our framework is trained with a multi-objective loss combining perceptual constraints, prior geometric consistency, and adversarial learning. The full loss function consists of four components:

$$\mathcal{L} = \lambda_{per}\mathcal{L}_{per} + \lambda_{r-per}\mathcal{L}_{r-per} + \lambda_{prior}\mathcal{L}_{prior} + \lambda_{adv}\mathcal{L}_{adv} \quad (5)$$

where λ_{per}, λ_{r-per}, λ_{struct} and λ_{adv} are balancing weights. Below we detail each component.

Perceptual Loss

The perceptual loss L_{per} minimizes the semantic discrepancy between generated image $\hat{I}_t$ and ground truth I_t in high-dimensional feature space:

$$\mathcal{L}_{per} = \sum_l \omega_l |\zeta_l(\hat{I}_t) - \zeta_l(I_t)|_1, \quad (6)$$

where $\zeta_l(\cdot)$ denotes features from the l-th layer of a pre-trained VGG-19 network [34], and ω_l is the layer-wise weight.

Regional Perceptual Loss

To enhance details of key regions, we compute perceptual loss specifically for face and hands:

$$\mathcal{L}_{r-per} = \sum_{r\in\mathcal{R}}\sum_l \omega_l |\zeta_l(\hat{I}_{t,r}) - \zeta_l(I^*_{t,r})|_1, \quad (7)$$

where $\mathcal{R} = \{face, lh, rh\}$, $\hat{I}_{t,r}$ is the generated regions, and $I^*_{t,r}$ is the corresponding grayscale patch from It.

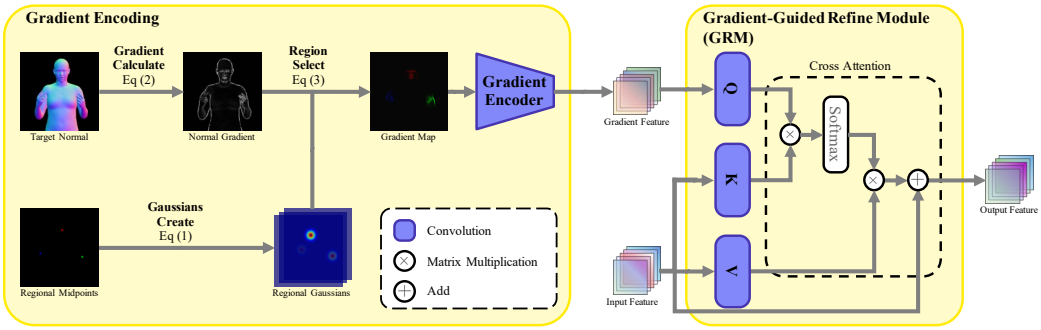


Figure 5. **Gradient-enhanced Refinement (GR)** begins with gradient encoding, where we compute the gradients of the input normal regions. These gradients are then filtered using Gaussians centered on the corresponding regions as weights to eliminate irrelevant areas. The processed gradients are encoded via a CNN and fed into the GRM as the Query for the cross-attention mechanism, aiming to refine key regions.

Prior Structure Loss

We enforce geometric consistency between the prior regions generated by RE and their corresponding target gradient:

$$\mathcal{L}_{struct} = \sum_{r \in \mathcal{R}} |R_r - I^*_{t,r}|_1. \tag{8}$$

This L1 loss directly constrains the structural alignment of key regions.

Regional Adversarial Loss

We design a multi-discriminator architecture with dedicated PatchGAN [] discriminators for key regions. The adversarial loss for generator G and discriminators is formulated as:

$$\mathcal{L}^G_{\text{adv}} = \sum_{r \in \mathcal{R}} \mathbb{E}\left[\log\left(1 - D_r\left(\hat{I}_{t,r}, G^r_t\right)\right)\right], \tag{9}$$

and for $\{D_r\}_{r \in \mathcal{R}}$ is formulated as:

$$\mathcal{L}^D_{\text{adv}} = \sum_{r \in \mathcal{R}} \Big(\mathbb{E}\left[\log D_r\left(I_{t,r}, G_{t,r}\right)\right] + \mathbb{E}\left[\log\left(1 - D_r\left(\hat{I}_{t,r}, \hat{G}_{t,r}\right)\right)\right] \Big) \tag{10}$$

where $\mathcal{R} = \{\text{face}, \text{lh}, \text{rh}\}$ denotes key regions, $D_r(\cdot)$ represents region-specific PatchGAN discriminator, $I_{t,r}$ and $\hat{I}_{t,r}$ are cropped regions from real/generated images, $G_{t,r}$ and $\hat{G}_{t,r}$ are cropped regions from $I_{t,r}$ and $\hat{I}_{t,r}$.

Unlike standard GANs [9] that use image-level discrimination, PatchGAN [12] outputs a matrix of patch-wise authenticity scores, enabling finer local detail discrimination. By incorporating gradient maps into the adversarial training, the model can better capture structural information and generate more realistic edges and fine details.

4. Experiments

4.1. Experimental Setup

Datasets. We adopt the largest and most widely adopted datasets in sign language production (SLP) research from SignAvatars [42]: the HamNoSys subset and the How2Sign [7] subset, these datasets provide anatomy-aligned SMPL-X annotations that precisely parameterize 3D body pose, hand articulation, and facial expressions.

- **HamNoSys subset**: This multilingual isolated-sign dataset integrates 5,800 video clips annotated with the HamNoSys notation system, covering four sign languages: Polish Sign Language (PJM) from the Polish SL corpus [17], and German (DGS), Greek (GRSL), and French Sign Language (LSF) from the DGS Corpus [21] and Dicta-Sign [19]. Since the original dataset was not divided, we have randomly divided it into a training set and a test set at an 8:2 ratio.
- **How2Sign**: A continuous American Sign Language (ASL) corpus containing 80 hours of video aligned with English transcripts recorded by 10 native signers. The official split includes 34k training segments, 1.5k validation, and 2k test samples. However, since only the annotations for the training set have been made public at this time, we have randomly re-divided it into a new training set and a test set at an 8:2 ratio.

Metrics. For evaluating the image quality of reconstructions, we adopted L1, Structural SIMilarity (SSIM)[40], and Learned Perceptual Image Patch Similarity (LPIPS)[44] following previous work. To specifically assess structural accuracy in key regions, we introduced region-level L1 (computed by cropping regions) and Average Keypoint Distance (AKD) [33]. For the video-based evaluation, we utilized Fréchet Video Distance (FVD)[**?**].

- Average Keypoint Distance (AKD) measuring Euclidean distances between normalized facial and hand keypoint coordinates.

Table 1. Quantitative evaluation results with SOTAs on the HamNoSys and How2Sign datasets. The best performance for each method is highlighted in **bold**.

(a) Results on HamNoSys dataset.

Dataset	HamNoSys					
Metric / Method	Image Level			Region Level		Video Level
	L1↓	SSIM↑	LPIPS↓	AKD↓	L1↓	FVD↓
NTED	0.02543	0.8461	0.04899	0.03488	0.010702	554.8±74.73
SignGAN	0.02938	0.8730	0.04723	0.02979	0.019519	582.4±45.93
PGMM	0.02886	0.8737	0.04504	0.02703	0.019700	454.2±34.78
GReg(Ours)	**0.01907**	**0.8836**	**0.03591**	**0.02429**	**0.009441**	**391.3±37.92**

(b) Results on How2Sign dataset.

Dataset	How2Sign					
Metric / Method	Image Level			Region Level		Video Level
	L1↓	SSIM↑	LPIPS↓	AKD↓	L1↓	FVD↓
NTED	0.02802	0.8375	0.05325	0.02342	0.05553	651.2±102.33
SignGAN	0.02843	0.8298	0.06643	0.02093	0.05093	613.7±83.68
PGMM	0.02725	0.8301	0.06354	0.01977	0.05132	564.0±88.02
GReg(Ours)	**0.02452**	**0.8557**	**0.05157**	**0.01726**	**0.04831**	**453.2±65.33**

- Fréchet Video Distance (FVD)[38]: It is a metric similar to FID[10] but specifically designed for assessing the quality of generated videos, where we employ a pre-trained I3D[4] model for feature extraction.

Implementation details. For flow estimation, we use the approach of [46] with a single occlusion mask, using ResNet-18 as the visual backbone to generate 50 key points in groups of 5 key points to generate 10 groups of TPS transforms. For geometry-aware generation, we use UNet architecture CNN with skip connections which use addition to combine features. For pose annotations, we used HRNet-W32 pre-trained on COCO-WholeBody[14].

For all datasets, we crop and resize input videos to 256*256. For each video, we randomly select a pair of source and driving images to train 100 epochs. For all experiments, we use the Adam optimizer with an initial learning rate of 2×10^{-4}, which is decreased by a factor of 10 at the 70th and 90th epochs. We train the entire network for 100 epochs on one NVIDIA GeForce 3090 GPU.

4.2. Comparison with state-of-the-art Methods

We compared our method with state-of-the-art methods, including the pose transfer method NTED [24], and two SLVG methods, SignGAN [31] and PGMM [32]. NTED is similar to our approach in terms of direction and uses SMPL-X as the input pose to validate our method's effect on key regions. SignGAN uses skeleton as input and employs adversarial training based on the coordinates of facial and hand key points. For PGMM, we replaced the real video with a heatmap as the input for flow estimation to verify the effects on different human representations.

We evaluated them on video reconstruction and cross-person video generation tasks. Video reconstruction is the task of reconstructing the original video by inputting one frame of the video as the source image and the rest of the frames as the driving images. Cross-person video generation is the task of transferring a sign language video from one person to another. It can effectively test the model's applicability to novel sign language video creation.

Video reconstruction. We evaluated the video reconstruction capabilities of all methods using the image quality, region quality, and video quality evaluation metrics mentioned earlier. As shown in Tab. 1, our proposed method surpasses the others, achieving state-of-the-art performance in terms of all metrics on three datasets.

Additionally, we visualized a set of reconstruction results, as shown in Fig. 6. NTED exhibited anatomical abnormalities in finger details, but it achieves better clarity compared to SignGAN and PGMM. This is because the normal map of SMPL-X provides better information on geometric edges and shapes. SignGAN and PGMM suffer from severe blurriness and distortion in the hands, which indicates that the abstract nature of skeleton and heatmap representations cannot directly express the geometric structure of the hands. Compared to these methods, our method excels in facial expressions, hand structure, finger geometric details, and semantic consistency between frames. The above analysis validates our method's ability to generate high-quality sign language videos by fully utilizing geometric information to refine key areas and produce anatomically



Figure 6. Qualitative Comparison with baselines on HamNoSys (left) and How2Sign (right) dataset. Our method demonstrates superior performance in generating facial expressions and hand structures compared to state-of-the-art approaches. Zoom in for the best view.



Figure 7. Qualitative comparison of ablation study. The qualitative comparison from our ablation study highlights that the SMPL-X significantly outperforms other representations. Additionally, the proposed modules in our framework demonstrate a remarkable improvement in enhancing hand structure and edge details.

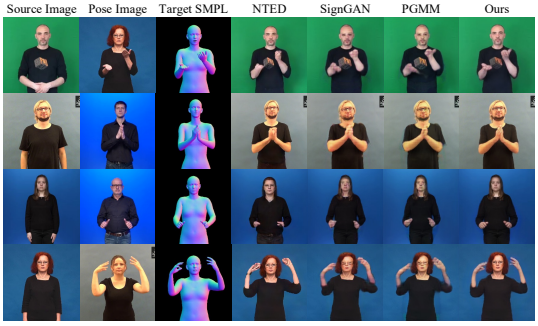


Figure 8. Qualitative comparison of cross-person video generation. Our method can generate more accurate facial expressions and hand structures than other methods.

accurate human representations.

Cross-person video generation. Fig. 8 shows the comparison results of our method with NTED, SignGAN, and PGMM in the cross-person video generation task. The images generated by NTED are often inconsistent with the identity and body shape in the source image. All baseline methods tend to produce images with incorrect finger structures. In contrast, our method can generate accurate facial expressions and hand structures while preserving the identity.

4.3. Ablation Study

We conducted ablation studies on the following four aspects: human representation, Region Prior Generation (RPG), Gradient-enhanced Refinement (GR), and adversarial training.

Effect of human representation. We compared three types of human representations: Skeleton, Heatmap, and SMPL-X (ours). We noticed that SMPL-X achieves slightly higher AKD values. However, despite the slightly higher value, SMPL-X produces visually superior results, as demonstrated in Fig. 7. This discrepancy arises because skeleton and heatmap representations, while effective for high-level feature extraction, lack the detailed structural information necessary for accurate sign language video generation. SMPL-X, on the other hand, provides precise geometric and anatomical details, enabling anatomically accurate and visually consistent outputs. This highlights the importance of using a representation that captures both high-level and fine-grained structural information for high-quality sign language video generation.

Effect of RPG. We evaluated the impact of RPG by: w/o

Table 2. Quantitative comparisons of ablation study on the HamNoSys dataset. The best performance for each method is highlighted in **bold**.

Method \ Metric	Image Level			Region Level		Video Level
	L1↓	SSIM↑	LPIPS↓	AKD↓	L1↓	FVD↓
For human representation						
Skeleton	0.02014	0.8736	0.03836	**0.02403**	0.010474	422.1±24.12
Heatmap	0.02098	0.8733	0.03789	0.02424	0.010108	412.3±37.23
For Region Prior Generation						
w/o RPG	0.02244	0.8636	0.04206	0.02804	0.011413	458.3±29.76
Conv	**0.01904**	0.8830	0.03610	0.02464	0.009556	401.3±39.71
Grayscale	0.01921	0.8813	0.03626	0.02504	0.009671	411.3±37.03
RGB	0.01910	0.8827	0.03632	0.02511	0.009654	410.3±38.62
For Gradient-Enhanced Refinement						
w/o GR	0.02044	0.8791	0.04129	0.02781	0.010238	441.4±40.11
Add	0.01944	0.8803	0.03806	0.02504	0.009676	411.5±37.38
Self-Att	0.01924	0.8822	0.03643	0.02449	0.009490	401.3±41.92
For adversarial training						
w/o adv	0.01967	0.8793	0.03687	0.02532	0.012665	434.6±42.63
Coords	0.01954	0.8801	0.03621	0.02511	0.011221	423.8±34.88
GReg(Full)	0.01907	**0.8836**	**0.03591**	0.02429	**0.009441**	**391.3±37.92**

RPG, Conv (Using convolutional layers in RE), Grayscale (Applying grayscale constraints to prior regions), RGB (Applying RGB). Our gradient-based RPG achieves the best performance, as shown in Tab. 2. The absence of RPG leads to significant performance degradation. This means RPG plays a crucial role in enhancing the quality of key regions by providing structural priors that complement the missing details in the original method. Gradient-based constraints outperform RGB and grayscale constraints because they emphasize edge and structural information, which is critical for generating anatomically accurate outputs.

Effect of GR. We evaluated the impact of GR by: w/o GR, Add (Adding gradient feature), Self-Att (Using self-attention of input feature). Our gradient-enhanced refinement achieves the best performance. This demonstrates the effectiveness of the GRM and the validity of using gradient features. GR enhances the global generation process by emphasizing structural information in key regions. Without GR, the improvements brought by RPG are significantly reduced, as the model fails to utilize the structural priors fully.

Effect of adversarial training. We evaluated the impact of adversarial training by: w/o adv, Coords (Using key point coordinates for adversarial training). Our gradient-based adversarial training achieves the best performance. Adversarial training with gradient information significantly enhances the structural details, particularly in edge regions. This is because gradient information provides stronger constraints on the geometric structure, leading to sharper and more accurate outputs. In contrast, using keypoint coordinates alone fails to capture the fine-grained details, resulting in blurred or distorted outputs.

5. Conclusion and Future Work

In this paper, we introduced a novel **G**eometry-Aware **Reg**ion Refinement (**GReg**) framework for sign language video generation. Unlike existing approaches, GReg fully exploits the rich geometric information provided by the SMPL-X to refine the structure of key regions, thereby generating sign language videos that are both structurally accurate and rich in detail. Additionally, we incorporated two novel methods: Regional Prior Generation (RPG) and Gradient-enhanced Refinement (GR). RPG focuses on modeling key regions such as the face and hands to generate local priors. GR, on the other hand, enhances global generation by leveraging gradient from key regions through the attention mechanism to refine details. By employing adversarial training on key regions and their gradient maps, we significantly enhance the realism of the generated videos.

Extensive qualitative and quantitative experiments demonstrated that GReg can generate sign language videos that are structurally accurate and detailed, showing substantial advantages over existing methods.

In the future, we plan to integrate diffusion models into the GReg framework to improve its generalizability and achieve coherent, detailed, and highly generalizable SLVG.

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